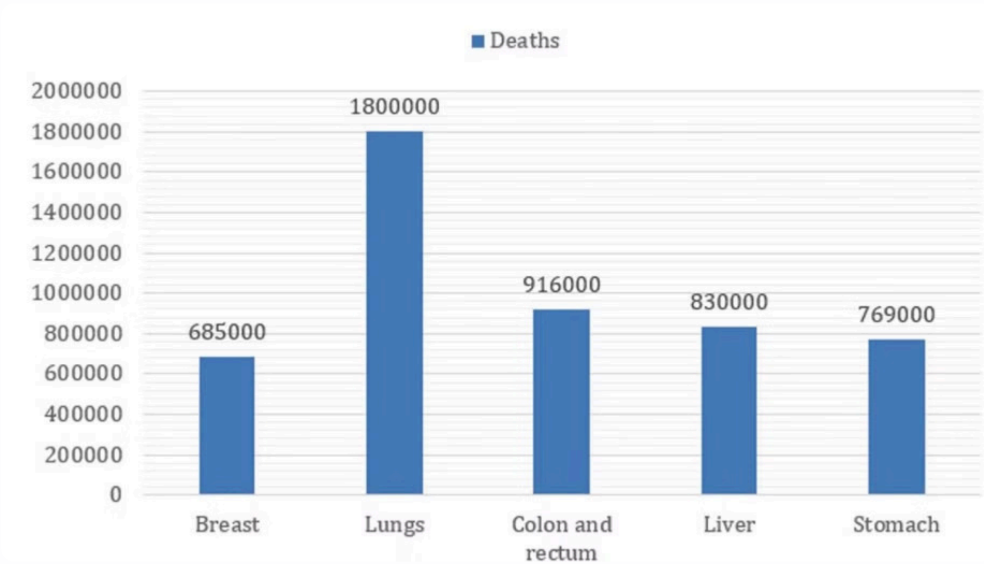
The Urgent Need

Global Challenge

1.8 million lung cancer deaths annually worldwide. Average diagnosis time: 3-6 months. 40% higher survival rate when detected early.



Global Distribution of Cancer-Related Deaths in 2020

Current Limitations

Manual review process, high risk of human error, limited specialist availability, growing patient volumes, resource constraints.

PulmoScan

PulmoScan is revolutionizing lung cancer detection through AI, making early diagnosis accessible, accurate, and efficient.

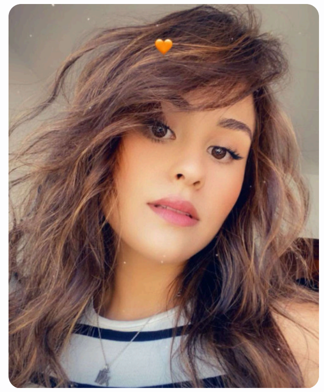
Because Every Breath Deserves a Future

Project Team & Contact Information

Project Leader:

Yosr Charrada

Email: yosr.chaarrada@esprit.tn



Project Manager:

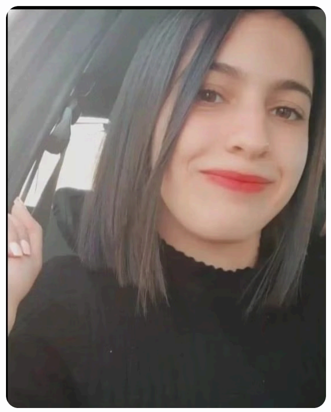
Maram Sliti

Email: maram.sliti@esprit.tn

Solution Architect:

Tessnim Etteib

Email: tessnim.etteib@esprit.tn



Data Scientist:

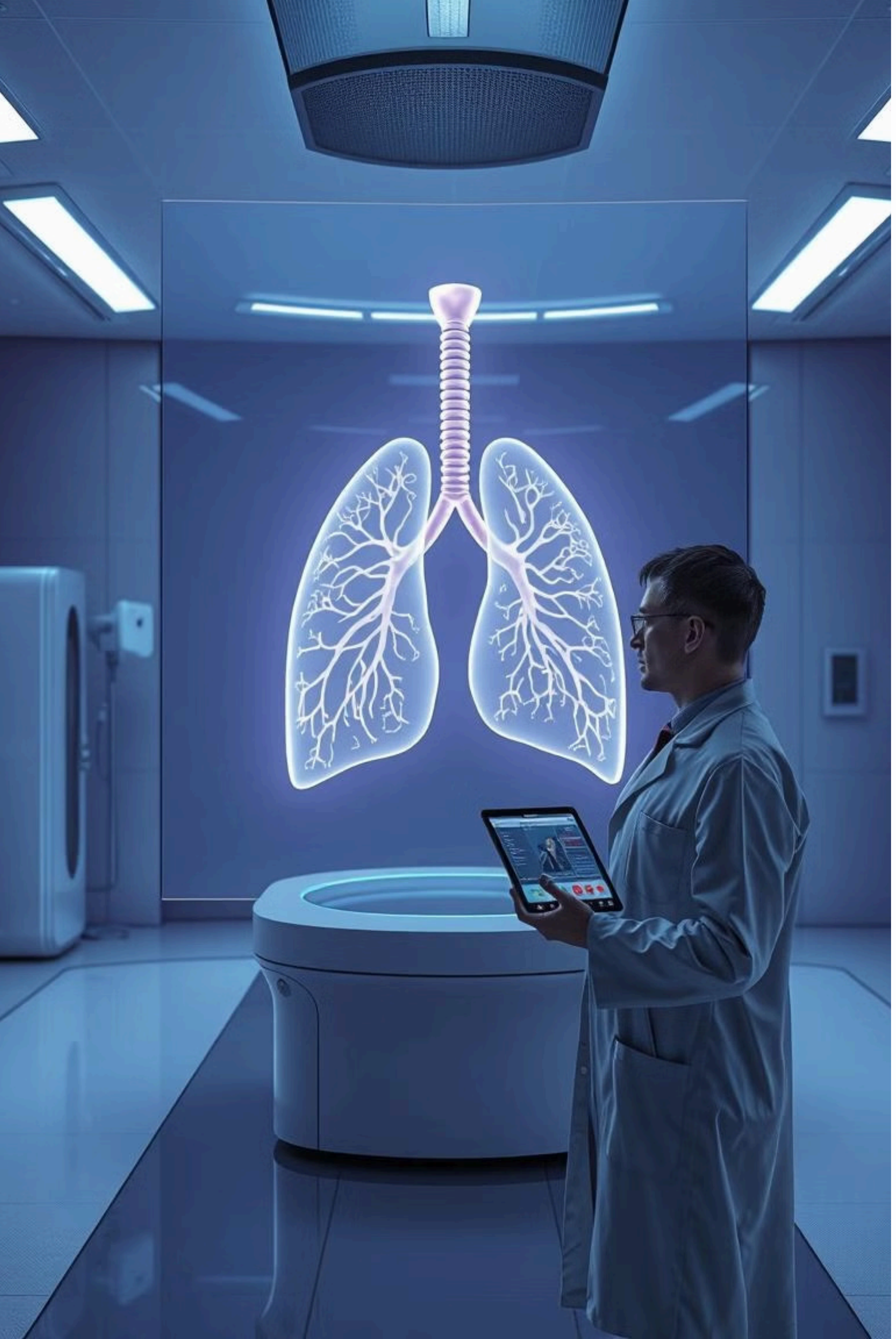
Nour Amori

Email: nour.amorri@esprit.tn

Data Scientist:

Meriem Mojaat

Email: mojaat.meriem@esprit.tn



Our Vision

1. **Speed**

   Rapid diagnosis, immediate treatment.

2. **Accuracy**

   AI-powered precision.

3. **Accessibility**

   Seamless integration, user-friendly.

4. **Impact**

   Saving lives, optimizing resources.

Value Proposition

Hospitals & Clinics

Faster diagnosis, reduced costs, improved resource use.

Medical Professionals

Automated screening, reduced workload, better decision support.

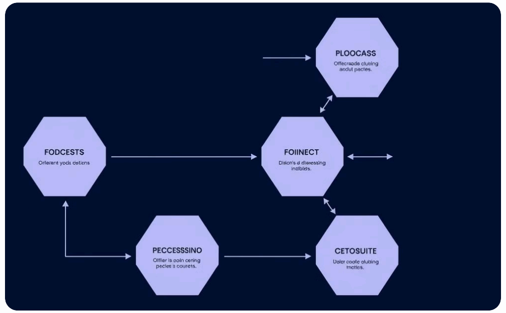
Healthcare Systems

Cost savings, increased efficiency, better outcomes.

Patients

Faster diagnosis, earlier treatment, improved survival.

Mastery of the Proposed Methodology



Structured Framework

Microsoft-Developed

Systematic Approach

Perfect for Healthcare

Key Stages of TDSP

1. Business Understanding
2. Data Acquisition & Understanding
3. Modeling & Development
4. Deployment & Integration
5. Customer Acceptance & Monitoring

Solution Overview

Automated Nodule Detection:

- *real-time scanning*
- *localize lung nodules*
- *measuring characteristiques*

Intelligent Classification:

normal tissue, benign, and malignant nodules

Cancer Type Identification:

- *Squamous cell carcinoma*
- *Adenocarcinoma*

Real-time Reporting:

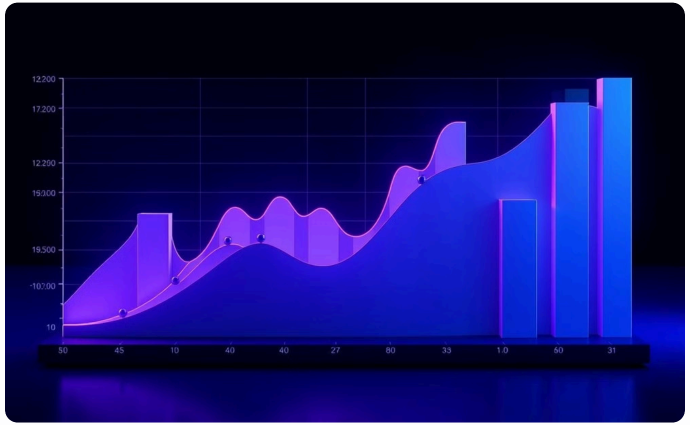
- *automated analysis*

Business & Data Science Goals

Business Goals	Data Science Goals
Make diagnoses more accurate by reducing human errors .	Reduce false positives and negatives using advanced AI.
Reduce doctors workload so they can focus on complex cases.	Automated nodule detection and classification .
Develop AI solutions for medical imaging and share findings through a scientific report.	Create adaptable AI models with clear documentation for researchers and doctors.
Make AI tools easy to use in hospitals and clinics.	Create a user-friendly system with visual outputs and reports.

Credibility Checker

Innovative Medical Imaging



Comprehensive Validation Framework

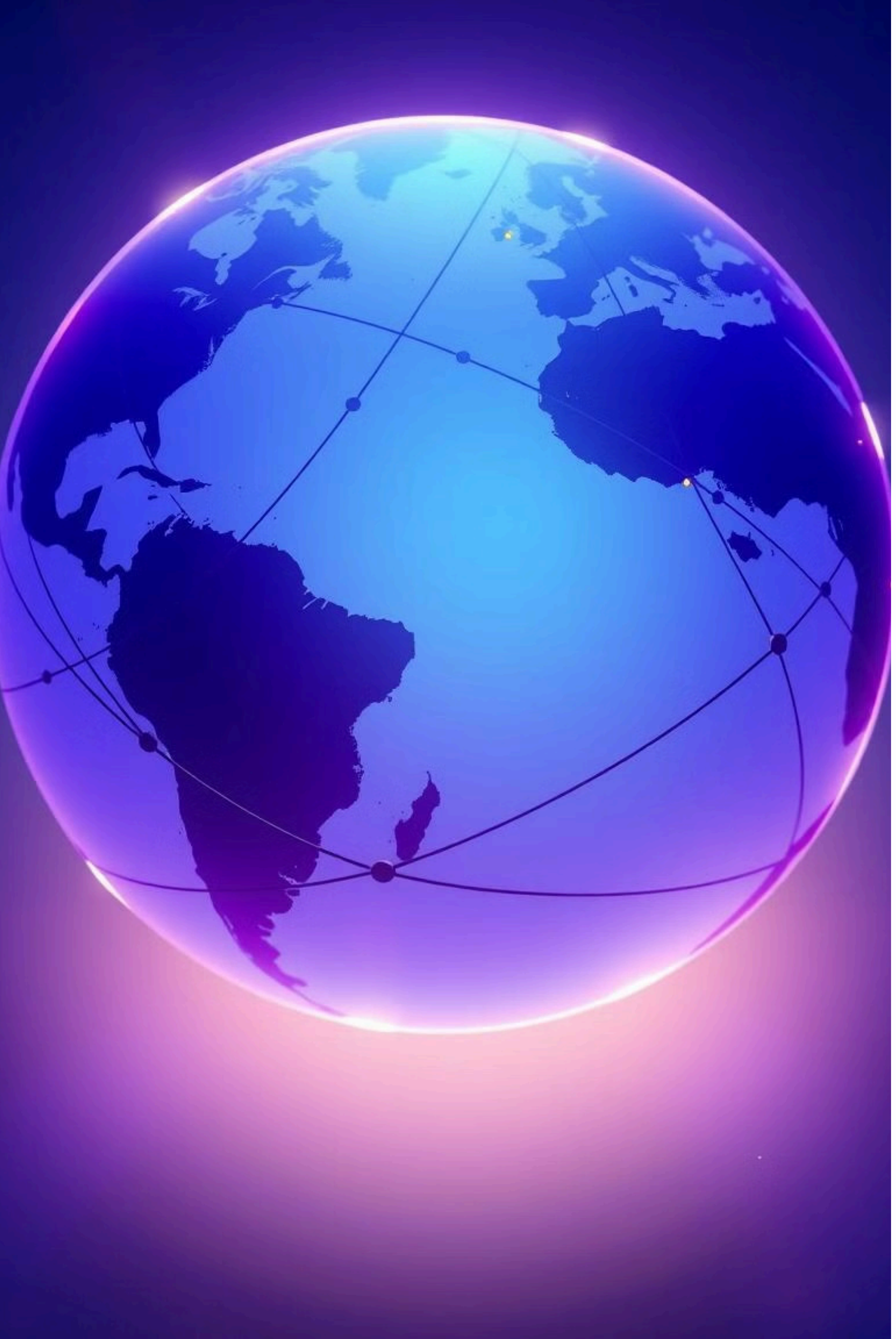
Rigorous testing ensures accuracy and reliability.

Multi-Dimensional Evaluation

Evaluated across multiple parameters for robustness.

Adaptive Learning Algorithms

Continuously improving for enhanced performance.



Sustainable Impact

1 SDG 3: Health & Well-being

Early cancer detection, improved survival rates, enhanced healthcare access, better patient outcomes.

2 SDG 9: Innovation

Advanced medical imaging, healthcare process optimization, technical innovation, system efficiency.

1

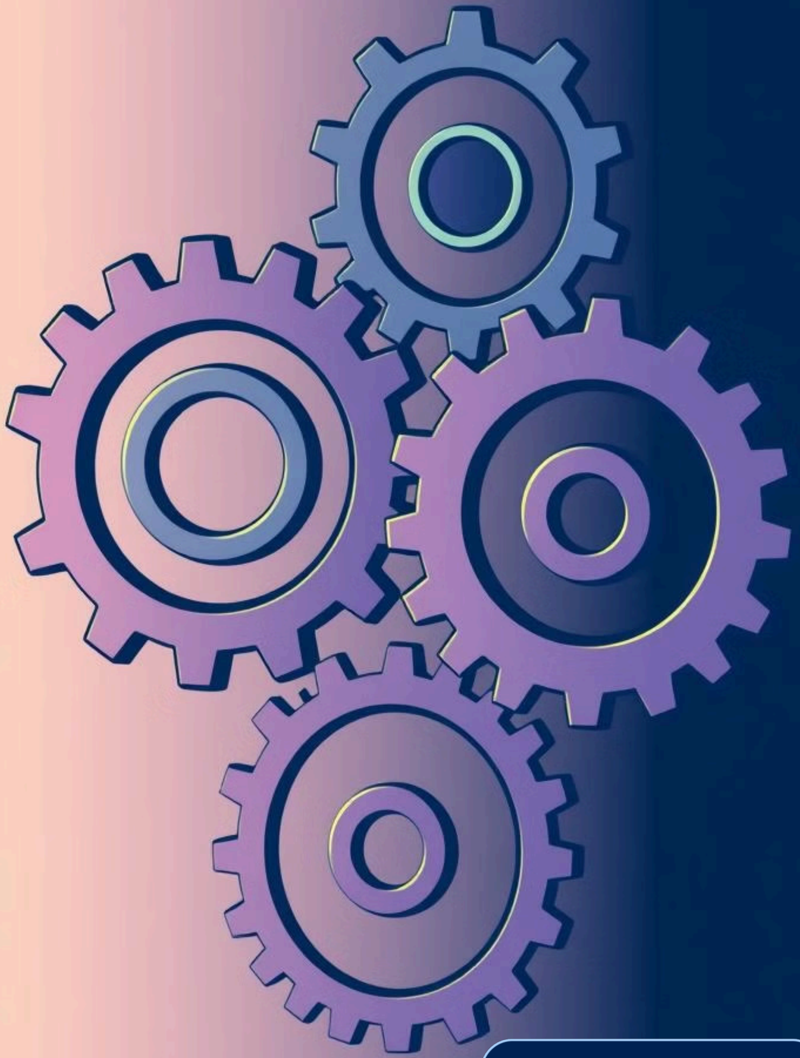
SDG 12: Responsible Production

Optimized AI efficiency, reduced resource consumption, sustainable operations, minimal environmental impact.

2

SDG 17: Partnerships

Healthcare collaboration, research partnerships, industry cooperation, knowledge sharing.



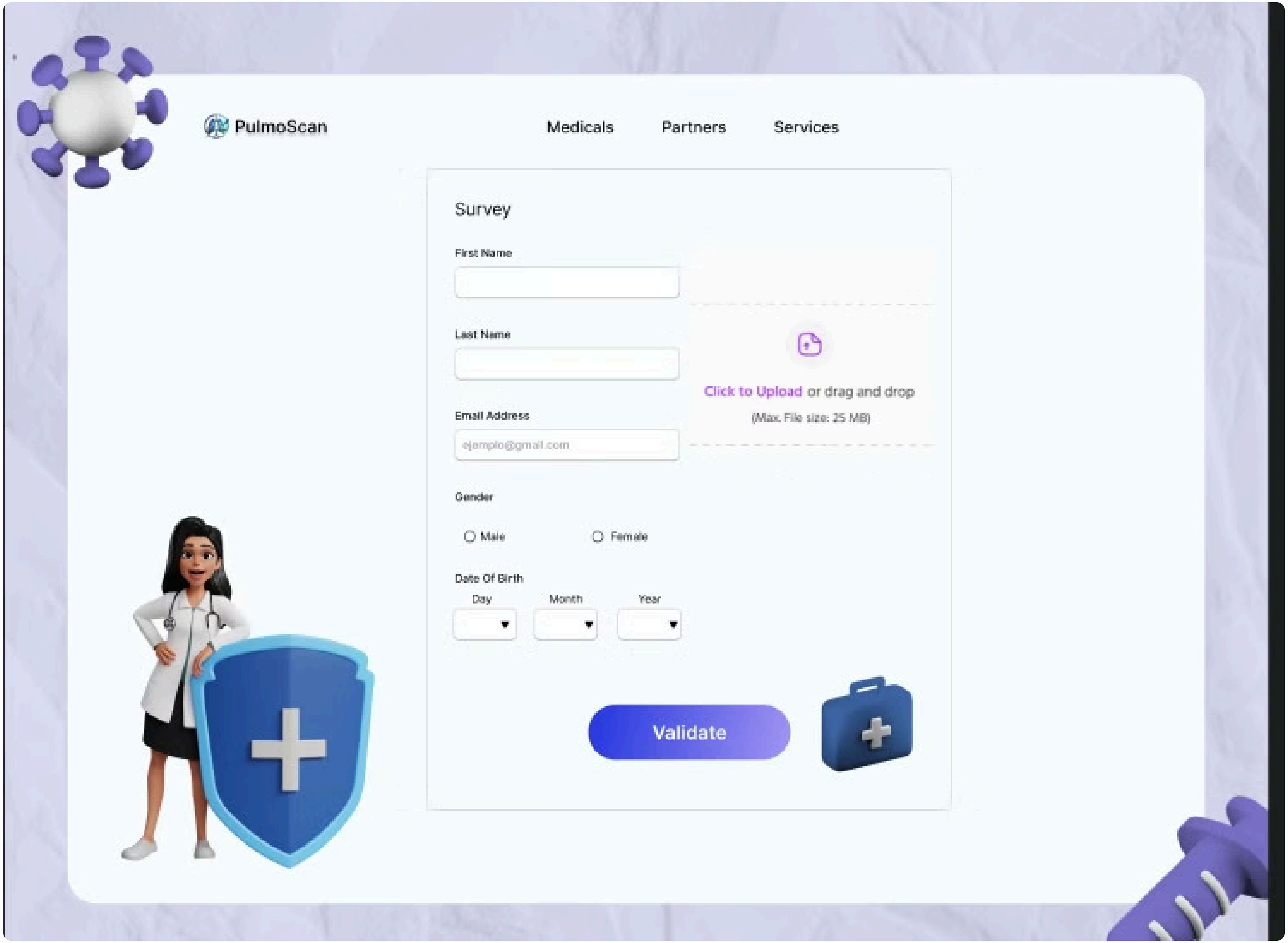
Stakeholder Benefits

1 Healthcare Providers

2 Patients

3 Healthcare Systems


PulmoScan
Medicals
Partners
Services
Sign Up
Revolutionizing Healthcare with AI
Enhancing Diagnosis & Treatment
Using AI-Powered Analysis, Patients Can Receive Accurate Medical Insights And Early Detection Of Lung Conditions.
Try Now
Detect, Diagnose, and Decide with AI-Powered Precision
AI-Powered Lung Health Detection with Accuracy and Efficiency
400k people online


PulmoScan
Medicals
Partners
Services
Survey
First Name
Last Name
Email Address
ejemplo@gmail.com
Click to Upload or drag and drop
(Max. File size: 25 MB)
Gender
Male
Female
Date Of Birth
Day
Month
Year
Validate

Success Metrics

Detection Accuracy	Classification Precision
Processing Time	Diagnosis Time Reduction

Risk Management

Technical Risks 1

Operational Risks 2

Mitigation Strategies 3

Conclusion

PulmoScan transforms lung disease diagnosis with AI, improving accuracy, efficiency, and patient outcomes, making it a valuable innovation in healthcare.